How Does Female Leadership Relate to Wage–Productivity Gaps in Colombia?

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Abstract

This study examines whether female leadership is associated with lower labor monopsony power in Colombian manufacturing firms. In labor markets with frictions such as search costs, limited mobility, and employer concentration, firms may exercise monopsony power and pay wages below workers' marginal productivity, generating wage-productivity gaps known as labor markdowns.

While these distortions arise from structural labor market frictions, firms differ substantially in how they exercise wage-setting power. Management decisions play an important role in shaping wage-setting practices and employment policies, suggesting that leadership characteristics may influence the extent to which firms exploit monopsony power. Prior research suggests that female leaders may adopt organizational practices that place greater weight on worker welfare, employment stability, and equitable compensation.

Using firm-level data from the Colombian Manufacturing Survey (2000-2019), I construct firm-level labor markdowns and relate them to the share of firm ownership held by women using regressions that control for firm and industry characteristics. Markdown estimates are derived from production-function methods that recover labor elasticities and compare the marginal revenue product of labor to wages.

The results show a negative association between female ownership shares and labor markdowns. These findings suggest that leadership composition may influence firms' wage-setting behavior and have implications for policies promoting greater female representation in firm leadership in emerging economies.

JEL Classification: D43, D63, J16,J42,J7, M12, M52

Keywords: female leadership, labor market power, wage–productivity gap, markdowns, monopsony, Colombia

1 Introduction

Standard microeconomic theory predicts that in competitive labor markets wages equal workers' marginal revenue of labor. Under this benchmark, firms act as wage takers, workers can freely move across jobs, and compensation accurately reflects productivity. Departures from this framework arise when firms face upward-sloping labor supply curves which gives them wage-setting power, also known as labor market monopsony (Robinson, 1969, cite class book). A growing literature shows that labor markets are often characterized by frictions such as search costs (Card, 2022; Azar et al., 2019; Lentz and Mortensen, 2010; Moscarini and Postel-Vinay, 2010; Manning, 2021), limited job mobility (Caldwell and Danieli, 2024; Schubert et al., 2025; Starr et al., 2021; McAdams, 2019), and employer concentration (Azar et al., 2017; Benmelech et al., 2018; Rinz, 2022). These frictions grant firms wage-setting power, allowing them to pay wages below workers' marginal productivity and generating a wage-productivity gap commonly referred to as a labor markdown.

Evidence suggests that such distortions are particularly pronounced in developing and emerging economies, where labor market frictions tend to be stronger. In Latin America and the Caribbean, workers receive roughly half of the value they generate on average, compared to about 65 percent in the United States and 81 percent in advanced economies (Alviarez et al., 2025). For Colombia, Amodio and de Roux, 2024 estimate a labor supply elasticity of 2.5, implying that workers produce approximately 40 percent more than they earn.

While monopsony power arises from structural features of labor markets, firms differ substantially in how they exercise wage-setting power. Management decisions shape wage-setting practices, organizational culture, and employment policies, suggesting that firm leadership may influence the extent to which monopsony power is exercised. This paper examines whether leadership characteristics—specifically gender composition—are systematically associated with differences in labor market power.

In particular, I investigate whether firms with greater female ownership exhibit lower labor markdowns than firms primarily owned by men. Prior research suggests that female leaders may adopt management styles that place greater weight on worker welfare and employment stability (Matsa and Miller, 2013; Tunyi et al., 2023), interpret productivity signals differently (Flabbi et al., 2017; Chen et al., 2019), or influence corporate decision-making in ways that reduce wage distortions (Levi et al., 2014; Flabbi et al., 2019). Moreover, labor market power has been shown to affect women more strongly than men (Sharma, 2023), raising the possibility that female leadership may mitigate gender-specific wage distortions within firms. If female leaders place relatively greater weight on worker welfare, long-term employment relationships, or equitable compensation, firms they lead may exercise monop-

sony power less aggressively, resulting in lower labor markdowns.

To examine this relationship, I use firm-level data from the Colombian Manufacturing Survey (EAM, from its Spanish initials) collected by the National Department of Statistics of Colombia (Departamento Administrativo Nacional de Estadística (DANE), n.d.). I estimate labor markdowns following the production-function methodology developed by Yeh et al., 2022. Specifically, markdowns are calculated using elasticities derived from a Cobb-Douglas revenue production function, with the wage bill used as the measure of labor input. Following Ackerberg et al., 2015, estimates of the revenue function elasticity allow the marginal revenue product of labor to be computed and compared to the average wage paid by the firm, yielding a firm-level wage markdown. This is the same methodology used by the Inter-American Development Bank in its Development in the Americas (DIA) 2025 report and the CompeteLAC dataset for markdown estimation. I then estimate regressions relating firm-level labor markdowns to the share of firm ownership held by women, controlling for observable firm and industry characteristics.

Preliminary evidence using industry-level markdowns estimated by Inter-American Development Bank, 2025 suggests a negative relationship between female ownership and labor markdowns, although the estimates are imprecise. Ongoing work will extend the analysis to firm-level markdowns and examine whether the relationship varies with labor market concentration and productivity differences across firms.

2 Literature Review

Table 1: Summary of Related Literature

Paper	Topic	Data	Main Finding
De Loecker, Eeckhout, & Unger (2020)	Labor market concentration	U.S. manufacturing	Production approach for a robust estimation of elasticities. Findings suggest markups rise not just to cover higher fixed costs, but also due to firms' market power.
Yeh, Macaluso & Hershbein (2022)	Labor markdown estimation	U.S. manufacturing	Most manufacturing plants operate in a monopsonistic environment. Average markdown 1.53 workers earning only 65 cents on the marginal dollar generated.
Amodio & de Roux (2024)	Colombia Labor market power	Colombian manufacturing plants (EAM)	Estimate a firm-level labor supply elasticity of around 2.5, implying that workers produce about 40% more than their wage level
Inter-American Development Bank (2025)	Latin-America and the Caribbean Market Power	Firm-Data LAC (For Colombia EAM)	On average, workers in Latin America and the Caribbean take home in the form of wages 50 percent of what they contribute to their employers.
Matsa & Miller (2013)	Female leadership	Nordic companies (Norway's 2006 quota)	Female leaders tend to emphasize employment stability and worker retention.
Flabbi et al. (2019)	Female executives	Italy employer-employee panel data	Female leadership influences wage-setting and reduce gender gaps within firms.
Levi, Li & Zhang (2014)	Female executives	Aquisition bids by S&P 1500 companies	Female directors less overestimate merger gains and help create shareholder value

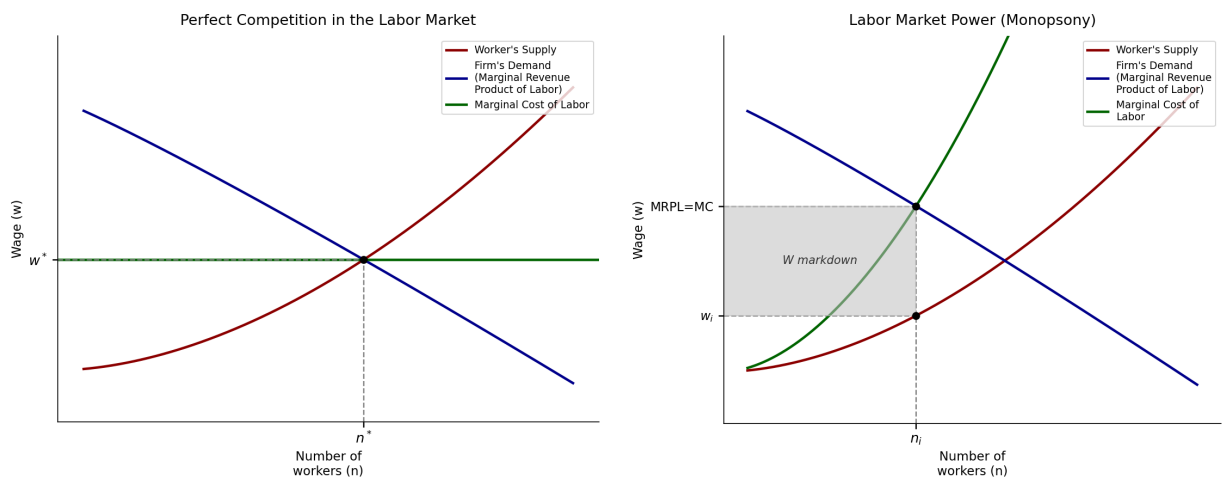
3 Theoretical Framework

To understand the wedge between wages and workers' marginal productivity, a useful starting point is the benchmark case of a competitive labor market. As shown in Panel (a) of Figure 1, under perfect competition the marginal cost of labor is constant and equal to the market wage, which each firm takes as given. Firms therefore hire labor up to the point where the marginal revenue product of labor (MRPL) equals the wage.

Under monopsony, by contrast, the firm faces an upward-sloping labor supply curve, as shown in Panel (b) of Figure 1. Hiring an additional worker requires raising the wage not only for the marginal worker but, when wages are set uniformly, for all inframarginal workers as well. As a result, the marginal cost of labor lies strictly above the labor supply curve. The profit-maximizing firm sets employment where MRPL = MCL, yielding both lower employment and a lower wage relative to the competitive benchmark. Crucially, at the optimum, workers are paid a wage w_i determined by the labor supply curve, yet their marginal contribution to firm revenue satisfies MRPL $> w_i$. The wedge between the marginal revenue product of labor and the wage—illustrated by the shaded region in Panel (b)—is the *markdown*, the central object of interest in this paper.

Figure 1: Labor Markets

(a) Perfect Competition (b) Monopsony



To understand how this markdowns arises, consider a firm's profit maximization problem with only one labor as input in a competitive market. The firm chooses labor input L_{it} to maximize profits:

$$\max_{L_{it}} \Pi_{it} = R_{it}(L_{it}) - \bar{w}L_{it}, \tag{1}$$

Where $R_{it}(L_{it})$ is the firm's revenue as a function of labor input $(P * Q(L))$, and $\bar{w}$ is the constant wage in the competitive benchmark that the firm takes as given. The first-order condition for profit maximization equates the marginal revenue($R'(L)$) with the marginal cost (W) such that

$$\frac{\partial R_{it}(L_{it})}{\partial L_{it}} = \bar{w} \tag{2}$$

For a monopsonistic firm, the wage is a function of the labor input $w_{it}(L_{it})L_{it}$ reflecting the upward-sloping labor supply curve observed in figure 1. The profit maximization problem becomes:

$$\max_{L_{it}} \Pi_{it} = R_{it}(L_{it}) - w_{it}(L_{it})L_{it}, \tag{3}$$

The first-order condition for profit maximization

$$\frac{\partial R_{it}(L_{it})}{\partial L_{it}} = \frac{\partial w_{it}(L_{it})}{\partial L_{it}} L_{it} + w_{it}(L_{it}) \tag{4}$$

Note that the monopolistic firms also maximizes profits by equating marginal revenue and marginal cost, but the marginal cost of labor is greater than the wage in this case. The left hand side of the equation is the marginal revenue product of labor $\text{MRPL}_{it} \equiv \frac{\partial R_{it}(L_{it})}{\partial L_{it}}$, while the right-hand side is the marginal cost of labor, which includes both the wage paid to the marginal worker and the increase in wages paid to inframarginal workers. Factoring out the wage $w_{it}(L_{it})$ We can rewrite the equation as:

$$\text{MRPL}_{it} = \left(\frac{\partial w_{it}(L_{it})}{\partial L_{it}} \frac{L_{it}}{w_{it}(L_{it})} + 1 \right) w_{it}(L_{it}) = \left(\varepsilon_L^{-1} + 1 \right) w_{it}(L_{it}) \tag{5}$$

Where the firm's perceived inverse elasticity of labor supply is defined as:

$$\varepsilon_L^{-1} \equiv \frac{\partial w_{it}(L_{it})}{\partial L_{it}} \frac{L_{it}}{w_{it}(L_{it})} \tag{6}$$

The markdown ν_{it} is then defined as the ratio of the marginal revenue product of labor to the wage:

$$\nu_{it} \equiv \frac{\text{MRPL}_{it}}{w_{it}(L_{it})} = \varepsilon_L^{-1} + 1, \tag{7}$$

Under this definition, $\nu_{it} = 1$ under perfect competition (infinite labor supply elasticity) and $\nu_{it} > 1$ whenever labor supply is upward-sloping.

4 Markdown Estimation - Production Approach

From the theoretical framework, we see that the key parameter governing the size of the markdown is the firm's perceived labor supply elasticity ε_L. Nonetheless, estimating a firm's perceived elasticity of labor supply in a general setting is challenging, in part because of the potential for firm market power over both inputs (monopsony) and output (monopoly) Yeh et al., 2022. To disentangle these two forces, I follow Yeh et al., 2022 and adopt the production approach of De Loecker et al., 2020.

To understand this production approach, let's consider a Cobb-Douglas production function[1] with three inputs — labor L_{it}, capital K_{it}, and materials M_{it} — in an environment where the firm holds power over both its output and labor markets. Following De Loecker et al., 2020, variable inputs (L_{it}, M_{it}) adjust frictionlessly within a period, while capital is subject to adjustment frictions.

The approach relies on the assumption that materials ($j = M_{it}$) is a fully flexible input satisfying three conditions: it incurs no adjustment costs within a period; its price is taken as given by the firm; and it is chosen statically, in the same period in which output is produced.

The firm's cost minimization problem is:

$$\mathcal{L} = r_{it}K_{it} + w_{it}(L_{it})L_{it} + p^m_{it}M_{it} - \lambda_{it}\left(P_{it}\bar{Q}_{it} - A_{it}K^{\theta^K}_{it}L^{\theta^L}_{it}M^{\theta^M}_{it}\right) \tag{8}$$

where A_{it} denotes firm-specific productivity[2] and θ^K, θ^L, θ^M are the output elasticities of capital, labor, and materials, respectively. The price of capital is r_{it}, the wage function is $w_{it}(L_{it})$, and the price of materials is p^m_{it}. The Lagrange multiplier λ_{it} captures the marginal cost of production.

The first-order conditions for M_{it} and L_{it} are:

$$\frac{\partial \mathcal{L}}{\partial M_{it}} = p^m_{it} - \lambda_{it}\varepsilon^M A_{it}K^{\theta^K}_{it}L^{\theta^L}_{it}M^{\theta^M-1}_{it} = 0. \tag{9}$$

[1]For the actual estimation, following De Loecker et al., 2020 I use a translog production function with polynomial terms as follows:

$$\log(GO)_{it} = \beta_0 + \beta_1\log(K)_{it} + \beta_2\log(L)_{it} + \beta_3\log(M)_{it} + \beta_4\log(K)^2_{it} + \beta_5\log(L)^2_{it} + \beta_6\log(M)^2_{it} + \beta_7\log(K)_{it}\log(L)_{it} + \beta_8\log(K)$$

[2]Following De Loecker et al., 2020, firm productivity is decomposed into a component observed by the firm but unobserved by the econometrician, modeled via an AR(1) process in which $A_{it} = A_{i,t-1} + \omega_t$. The component observed by the firm is ω_t, the innovation in productivity.

$$\frac{\partial \mathcal{L}}{\partial L_{it}} = \frac{\partial w_{it}(L_{it})}{\partial L_{it}} L_{it} + w_{it}(L_{it}) - \lambda_{it} \varepsilon^L A_{it} K_{it}^{\theta^K} L_{it}^{\theta^L - 1} M_{it}^{\theta^M} = 0, \tag{10}$$

We know from the theoterical framework that $\frac{\partial w_{it}(L_{it})}{\partial L_{it}} L_{it} + w_{it}(L_{it}) = \left(\varepsilon_L^{-1} + 1\right) w_{it}(L_{it})$. Also using the production function $Q_{it} = A_{it} K_{it}^{\theta^K} L_{it}^{\theta^L} M_{it}^{\theta^M}$, we can simplify the first-order conditions to:

$$p_{it}^m = \lambda_{it} \varepsilon^M \frac{Q_{it}}{M_{it}} \tag{11}$$

$$\left(\varepsilon_L^{-1} + 1\right) w_{it}(L_{it}) = \lambda_{it} \varepsilon^L \frac{Q_{it}}{L_{it}} \tag{12}$$

Since λ_{it} is the Lagrange multiplier on the production constraint, interpreted as marginal cost, the firm's markup—the price-to-marginal-cost ratio—is therefore:

$$\mu_{it} = \frac{P_{it}}{\lambda_{it}}. \tag{13}$$

Defining the revenue shares of labor and materials as:

$$\alpha_{it}^L \equiv \frac{w_{it}(L_{it}) L_{it}}{P_{it} Q_{it}}, \qquad \alpha_{it}^M \equiv \frac{p_{it}^m M_{it}}{P_{it} Q_{it}}, \tag{14}$$

substituting into the first-order condition for materials yields a markup estimator:

$$\mu_{it} = \frac{\theta^M}{\alpha_{it}^M}, \tag{15}$$

Substituting analogously into the first-order condition for labor gives:

$$\left(\varepsilon_L^{-1} + 1\right) = \frac{\theta^L}{\alpha_{it}^L} \frac{1}{\mu_{it}}, \tag{16}$$

Recalling from the theoretical framework that the markdown is defined as:

$$\nu_{it} \equiv \frac{\text{MRPL}_{it}}{w_{it}(L_{it})} = \varepsilon_L^{-1} + 1, \tag{17}$$

the markdown can be expressed in terms of observable quantities as:

$$\nu_{it} = \frac{\theta^L}{\alpha_{it}^L} \cdot \frac{1}{\mu_{it}} = \frac{\theta^L}{\alpha_{it}^L} \cdot \frac{\alpha_{it}^M}{\theta^M} = \frac{\theta^L}{\theta^M} \cdot \frac{p_{it}^m M_{it}}{w_{it}(L_{it}) L_{it}}. \tag{18}$$

This expression shows that the markdown can be estimated using the ratio of the firm's expenditure on materials to its wage bill, scaled by the ratio of output elasticities. The

output elasticities θ^L and θ^M are estimated from the production function, while the input shares α_{it}^L and α_{it}^M are computed from observed data on wages, materials, output, and prices. This approach allows for the estimation of firm-level markdowns while controlling for both output and labor market power.

5 Data

This paper uses firm-level data from the Annual Manufacturing Survey (*Encuesta Anual Manufacturera*, EAM) conducted by Colombia's National Administrative Department of Statistics (DANE). The EAM is a census of medium and large manufacturing establishments that provides detailed and standardized information on firms' production, input use, employment composition, and financial accounts. The original dataset spans 1995–2024 and covers approximately 7,000 manufacturing establishments per year. I restrict the analysis to the period 2000–2019 to ensure data consistency and to avoid disruptions associated with the COVID-19 pandemic. Although the data are collected at the establishment level, I aggregate to the firm level by summing across all plants belonging to the same firm in a given year, using the unique firm identifier provided in the survey.

The resulting dataset is an unbalanced panel that reflects firm entry and exit over time. Following DANE's technical guidelines, the sample is restricted to firms with at least 10 employees, which ensures consistency in reporting standards and improves the reliability of firm-level measures. Observations with missing or implausible values—such as negative output or extreme input ratios—are excluded, affecting approximately 0.2 percent of total observations.

Main Explanatory Variable. The main explanatory variable is the firm-level share of female ownership, constructed from variables recording the number of male owners (`c4r4c1t`) and female owners (`c4r4c2t`) reported in the survey.[3] The female ownership share is defined as:

$$\text{Female Ownership Share}_{it} = \frac{\text{Number of female owners}_{it}}{\text{Total number of owners}_{it}}. \tag{19}$$

This variable ranges from 0 to 1 and captures the gender composition of firm ownership. Ownership is measured annually and may vary over time; observations with zero reported

[3]In the form, these variables represent the total number of owners, partners, and unpaid family members. They are disaggregated by gender as well as by occupational category—professionals; technicians and technologists; production workers (foreign and local); operators and producers; and managers and sales personnel. I use the sum of all owner types to construct the total ownership variable.

owners are excluded from the analysis. One limitation of this variable is that a nonzero ownership count is reported for only approximately 30 percent of firm-year observations, which may reflect genuine non-reporting or survey design issues. Since the analysis is restricted to observations with nonzero ownership counts, the results should be interpreted with caution, as firms with missing ownership data may differ systematically from those for which ownership is observed. I discuss potential robustness checks to address this concern in the empirical section of this paper.

Main Outcome Variable. The main outcome variable is the firm-level labor markdown, defined as the ratio of a worker's marginal revenue product of labor to the wage paid by the firm. Markdown estimates are derived from the production function framework described in Section 4, which combines estimated output elasticities with observed input cost shares.[4]

The dataset provides the inputs required to implement the production function methodology of De Loecker et al., 2020. Labor is measured as total employment, defined as the sum of permanent employees, firm owners, and temporary workers, both directly hired and outsourced. The wage bill corresponds to total remuneration paid to permanent and temporary employees. Capital is proxied by the book value of fixed assets, with its cost approximated using a user-cost formula that incorporates depreciation rates, interest rates, and inflation. Materials are measured as total expenditure on raw materials, intermediate inputs, and packaging; consistent with the requirements of the proxy-variable estimator, materials are treated as the flexible input in production. Gross output is proxied by firm level total sales at factory prices, net of direct taxes, as reported in the survey. The price of materials is calculated as the ratio of total materials expenditure to the quantity of materials used.

Outlier Treatment and Sample Restrictions. Administrative firm-level data frequently contain extreme values that arise from measurement error, coding mistakes, or genuine but highly influential outliers. Left uncorrected, such values can severely bias the production-function estimates that underpin the markdown calculations. Following Yeh et al., 2022, I apply three filters. First, I winsorize key variables—output, labor, capital, and materials—at the top and bottom 1 percent of their respective distributions to mitigate the influence of extreme observations. Second, I exclude observations with non-positive values for output or

[4] I improved upon De Loecker et al., 2020 estimation procedure by using a Nelder-Mead primary optimizer, since the GMM surface can be non-smooth and non-convex, especially for the translog specification with its many cross-terms. Rvmmin (Variable metric nonlinear function minimization) provides a gradient-based refinement when Nelder-Mead stalls near the minimum. The estimation is run in a loop iterating over every distinct 4-digit ISIC sector in the data, with elasticities recovered at the firm level. For firms where the model did not converge, I follow Ackerberg et al., 2015 and use a Cobb-Douglas specification at the sector level, imputing the median sector-specific elasticities.

any input, as these are inconsistent with the production framework. Third, to ensure that materials satisfy the flexible-input assumptions required for identification, I compute each firm's materials cost share—defined as materials expenditure as a share of total costs—and exclude observations whose materials cost share falls below one quarter of the sector-year median. Plants with anomalously low materials cost shares are unlikely to satisfy the conditions of no adjustment costs, static input choice, and the absence of input market power, and their inclusion could contaminate the proxy-variable estimator.

Descriptive Statistics. The final dataset comprises of an unbalanced panel of 6,323 Colombian manufacturing firms observed over the period 2000–2019, yielding 29,942 firm-year observations spanning 92 industries and 23 geographic regions. Table 2 presents descriptive statistics for the key variables. At the firm level, the average number of female owners is 0.67, with a mean female ownership share of 35.3 percent. Firms employ an average of 34 workers, with annual wages per employee averaging $1.83 thousand USD and mean gross output of $1.22 millionUSD; all of these variables exhibit substantial dispersion, reflecting the heterogeneity in firm size and performance that is typical of manufacturing surveys of this type.

The estimated production-function parameters are broadly consistent with the existing literature and lend credibility to the markdown estimates. The mean output elasticity of labor is 0.36, with a median of 0.39, while the output elasticity of materials averages 0.62 and the elasticity of capital averages 0.08. These figures are in line with estimates reported for other manufacturing settings using the same methodology De Loecker et al., 2020; Yeh et al., 2022. The dominance of materials over labor in the cost structure is consistent with the capital-light, assembly-intensive nature of much Colombian manufacturing, which supports the use of materials as the flexible input. The mean product markup is 1.23, implying that the average firm prices output approximately 23 percent above marginal cost; the median markup of 1.31 falls within the range estimated by De Loecker et al., 2020 for U.S. manufacturing and by Inter-American Development Bank, 2025 for the broader Latin American and Caribbean region, where the average manufacturing markup is reported to be 87 percent above marginal cost. Turning to the markdown estimates, the median firm-level markdown is 1.93, implying that the median worker produces nearly twice as much as they are paid—or equivalently, receives approximately 52 cents on each dollar generated at the margin. The mean inverse markdown of 0.59 indicates that, on average, workers receive about 59 percent of their marginal revenue product, a figure that is close to the 50 percent reported for Latin America and the Caribbean by the Inter-American Development Bank Inter-American Development Bank, 2025 and to the 65 percent implied by the average markdown of 1.53

estimated for U.S. manufacturing by Yeh et al., 2022. These estimates also align with the findings of Amodio and de Roux, 2024 for Colombia, who report a labor supply elasticity of approximately 2.5—implying workers produce about 40 percent more than their wage—a degree of monopsony broadly consistent with the markdown distribution documented here. The substantial dispersion in markdowns across firms, with the interquartile range spanning from 1.04 to 3.41, is also consistent with the heterogeneity documented in the literature Yeh et al., 2022 and motivates the firm-level regression analysis.

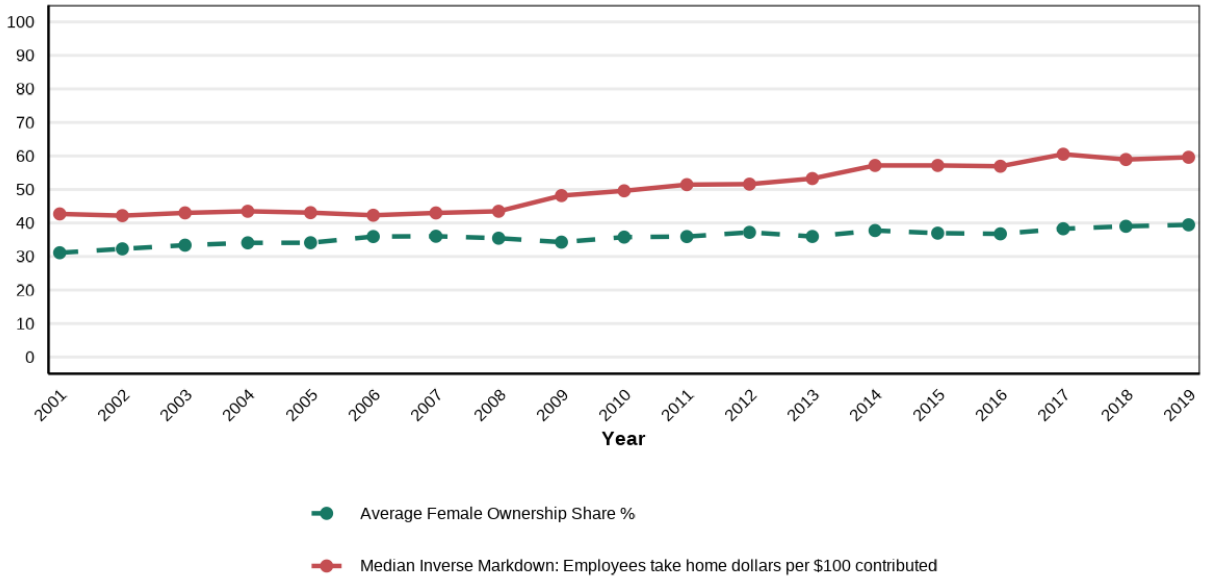
Table 2: Descriptive Statistics

Overall				
Variable	*Value*			
Observations	29,942			
Years	20			
Firms	6,323			
Geographies	23			
Industries	92			
By Firm				
Variable	*Mean*	*Min*	*Max*	*Std. Dev.*
Female owners (N)	0.667	0	58	0.900
Male owners (N)	1.054	0	30	0.809
Female ownership share (%)	35.319	0	100	37.579
Employees (N)	34	1	4,025	99
Wage per employee ($ thousands USD)	1.827	0	349.180	2.296
Gross output ($ millions USD)	1.219	0	756.959	11.356
By Firm – Estimation Parameters				
Variable	*Mean*	*p25*	*Median*	*p75*
Output elasticity of labor	0.358	0.239	0.386	0.433
Output elasticity of capital	0.081	0.033	0.064	0.106
Output elasticity of materials	0.624	0.528	0.641	0.693
Total Factor Productivity (TFP)	0.500	2.879	3.494	4.402
Labor output share	0.150	0.072	0.122	0.193
Materials output share	0.479	0.336	0.474	0.624
Capital output share	0.073	0.019	0.042	0.087
Markup	1.231	1.026	1.310	1.736
Markdown	1.059	1.043	1.933	3.406
Inverse markdown	0.594	0.294	0.517	0.959

Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019. Note: The wage variable corresponds to the total wage bill of the firm divided by the number of employees, while gross output is proxied by total sales at factory price without direct taxes. The dollar values are calculated using an exchange rate of 1 USD = 3,670 COP. Output elasticities and input shares are estimated using the production function approach by De Loecker et al., 2020. Markup and markdown estimates follow Yeh et al., 2022.

In Figure 2, I present how the inverse markdown and female ownership share have evolved over time. The inverse markdown is defined as $\nu_{it}^{-1} = w_{it}(L_{it})/\text{MRPL}_{it}$, so that higher values correspond to lower degrees of monopsony power, ranging from 0 to 100 where 100 indicates perfect competition. Over the sample period, both series trend modestly upward: the inverse markdown rises from approximately 42.7 cents per dollar of worker contribution in 2000 to 59.6 cents in 2019, while the female ownership share increases from 31.1 percent to 39.5 percent. This co-movement offers preliminary suggestive evidence of a positive association between female ownership and worker compensation relative to productivity. The upward trend in the inverse markdown is also consistent with the increased competitive pressure that Colombian manufacturing firms faced following free trade agreements signed with the United States and the European Union during the 2000s, which may have reduced firms' ability to exercise monopsony power over workers.

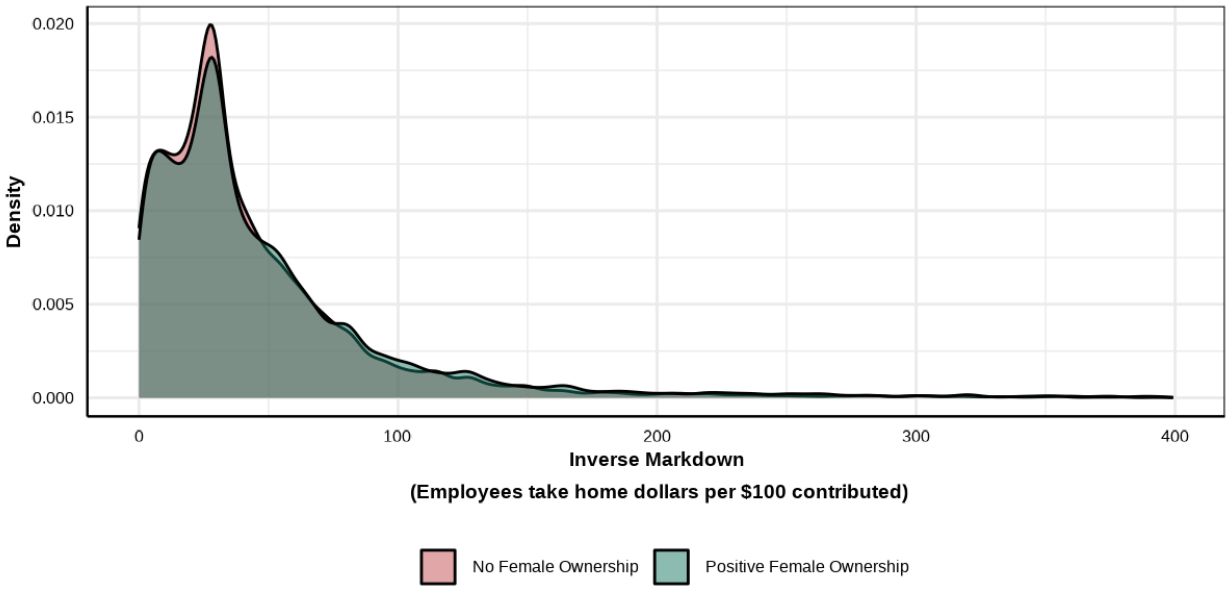
Figure 2: Inverse Markdown and Female Ownership Share Overtime



Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019.

In Figure 3, I present the distribution of the inverse markdown separately for firms with no female ownership and firms with at least one female owner. Firms without any female ownership exhibit a distribution more concentrated at lower values of the inverse markdown, suggesting that they tend to pay workers a larger share below their marginal productivity compared to firms with at least partial female ownership.

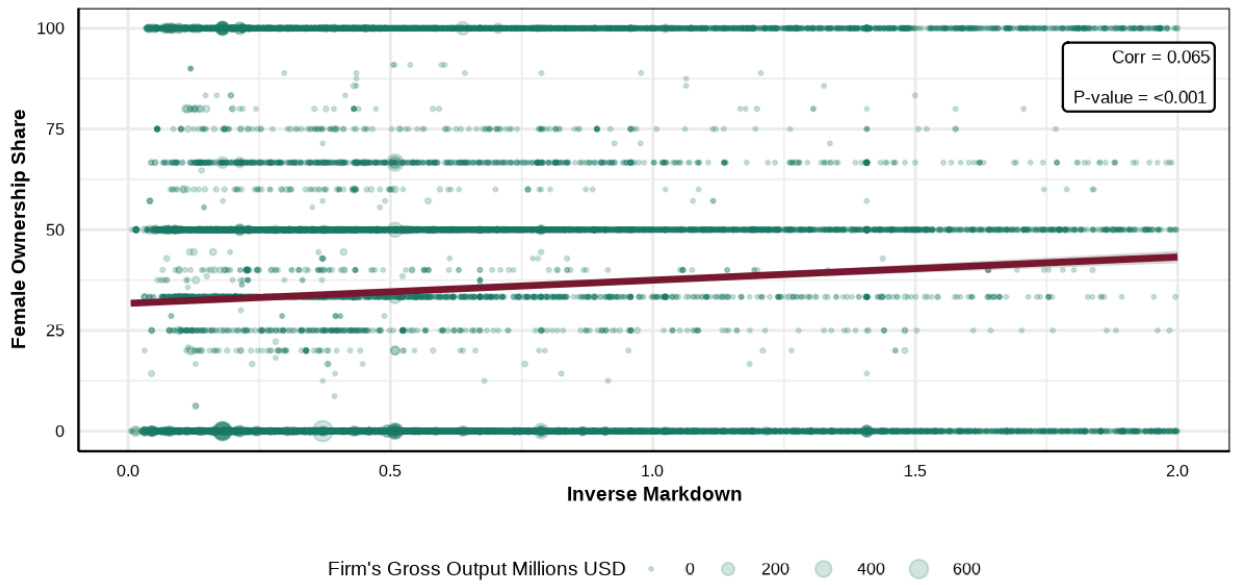
Figure 3: Distribution of Inverse Markdown by Female Ownership



Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019.

This result is consistent with Figure 4 and provides preliminary evidence of a positive association between the share of female ownership and the inverse markdown, suggesting that firms with greater female ownership tend to pay workers a higher share of their marginal revenue product. However, these descriptive patterns are purely correlational and do not account for potential confounding factors that may drive both female ownership and markdowns, such as industry composition, firm size, or regional labor market conditions. The regression analysis in the next section will provide a more rigorous examination of this relationship while controlling for these factors.

Figure 4: Correlation plot of Inverse Markdown and Female Share of Ownership



Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019.

6 Empirical Framework

To examine the relationship between female ownership and labor market power, I estimate the following linear regression:

$$\nu_{it}^{-1} = \beta \, \text{Female}_{it} + \mathbf{X}_{it}'\boldsymbol{\gamma} + \alpha_t + \alpha_s + \alpha_r + \varepsilon_{it}, \tag{20}$$

where ν_{it}^{-1} is the inverse markdown for firm i in year t. The outcome is expressed in levels so that β has the interpretation of a absolute difference in what employees receive in terms of their marginal revenue product between female-owned and male-owned firms, conditional on controls.

The main variable of interest, Female_{it}, measures the share of female ownership of the firm. The coefficient β therefore captures the conditional association between female ownership and the firm's degree of monopsony power. I expect β to be positive based on the theoretical and empirical literature suggesting that female-owned firms may be more likely to prioritize worker welfare and pay higher wages relative to productivity, which would correspond to a lower markdown and thus a higher inverse markdown.

The vector $\mathbf{X}_{it}$ includes a set of firm-level controls. Firm age accounts for the possibility that older firms have had longer to build labor market power through reputation or search frictions. Firm size, measured by gross output, controls for scale effects on both the

production side and the firm's capacity to set wages. Finally, I include the share of female workers and the share of skilled workers in the firm's workforce to account for differences in employment composition that may independently affect markdowns.

The specification includes three sets of fixed effects. Year fixed effects α_t absorb common macroeconomic shocks and aggregate trends in markdowns over time. Sector fixed effects α_s account for permanent differences across industries in production technology, product market structure, and average labor market competitiveness. Region fixed effects α_r control for local labor market conditions, including differences in the outside options available to workers. Standard errors are robust to account for heteroskedasticity.

7 Preliminary Results

Table 3: Regression Results

	(1)	(2)	(3)
`own_w_share`	0.066***	0.010*	0.012**
	(0.007)	(0.006)	(0.006)
Fixed Effects	Year	Year-Industry	Year-Industry & Geography

Note: *p<0.10; **p<0.05; ***p<0.01. Standard errors in parentheses.

Table 3 presents the results of the regression analysis. In the baseline specification (Column 1), which includes only year fixed effects, the coefficient on the female ownership share is positive and statistically significant at the 1 percent level, suggesting that a higher share of female ownership is associated with a higher inverse markdown, or equivalently, a lower degree of monopsony power. Specifically, a 10 percentage point increase in the share of female ownership is associated with a 0.66 percentage point increase in the inverse markdown, implying that for every 100 dollars of marginal revenue product, workers receive approximately 0.66 dollars more in wages.

When industry fixed effects are added in Column 2, the coefficient on female ownership share remains positive but its magnitude decreases substantially to 0.010 and is only significant at the 10 percent level. This suggests that part of the association observed in the baseline may be driven by industry composition, as certain industries with higher female ownership may also have more competitive labor markets. Finally, in Column 3, which includes both industry and geography fixed effects, the coefficient on female ownership share remains positive and becomes statistically significant at the 5 percent level, with a magnitude of 0.012. This indicates that even after controlling for both industry and regional factors,

there is still a positive association between female ownership and the inverse markdown, although the effect size is smaller than in the baseline specification.

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